Pandas Library

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Content

- Introduction
- DataFrame
- Operations in DataFrame

Pandas

- https://pandas.pydata.org/
- Use open source data analysis and manipulation tool
  - Fast
  - Powerful
  - Flexible
  - Easy

DataFrame

- Declare a data frame as columns

```
import pandas as pd

df = pd.DataFrame({
    'X': [13,  30, 'A'],
    'Y': [15,  32, 'B'],
    'Z': [10, 29, 'O'],
    'T': [12, 28, 'AB']},
    index = [1, 2, 3]
)
```

	X	Y	Z	T
1	13	15	10	12
2	30	32	29	28
3	A	B	O	AB

DataFrame

- Declare a data frame as rows

```
import pandas as pd

df = pd.DataFrame({
 data=[[1, 2, 3],
[4, 5, 6], [7, 8, 9]],
 columns=['X', 'Y', 'Z'],
 index=[1, 2, 3]
)
```

	X	Y	Z
1	1	2	3
2	4	5	6
3	7	8	9

DataFrame

- Read/Write from csv files (read more for Excel, SQL Query, Database, etc.)

```
df = pd.read_csv('sample_data/california_housing_train.csv')
df.to_csv('dataframe.csv')
print(df.head(5))
```

```
   longitude  latitude  ...  median_income  median_house_value
0    -114.31     34.19  ...         1.4936             66900.0
1    -114.47     34.40  ...         1.8200             80100.0
2    -114.56     33.69  ...         1.6509             85700.0
3    -114.57     33.64  ...         3.1917             73400.0
4    -114.57     33.57  ...         1.9250             65500.0

[5 rows x 9 columns]
```

Concepts

column

index

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	66900.0
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	80100.0
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	85700.0
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	73400.0
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	65500.0

observation

Indexing

```python
print(df.loc[0])
```

```
longitude                -114.3100
latitude                   34.1900
housing_median_age         15.0000
total_rooms              5612.0000
total_bedrooms           1283.0000
population               1015.0000
households                472.0000
median_income               1.4936
median_house_value      66900.0000
Name: 0, dtype: float64
```

```python
print(df['median_house_value'])
```

```
0    66900.0
1    80100.0
2    85700.0
3    73400.0
4    65500.0
Name: median_house_value, dtype: float64
```

```python
print(df.loc[0, 'median_house_value'])
```

```
66900.0
```

Pivot

[index, columns] combination must be unique

```
df.pivot(
 index='housing_median_age',
 columns='total_rooms',
 values='median_house_value'
)
```

total_rooms	720.0	1454.0	1501.0	5612.0	7650.0
housing_median_age					
14.0	NaN	NaN	73400.0	NaN	NaN
15.0	NaN	NaN	NaN	66900.0	NaN
17.0	85700.0	NaN	NaN	NaN	NaN
19.0	NaN	NaN	NaN	NaN	80100.0
20.0	NaN	65500.0	NaN	NaN	NaN

Pivot

```
df.pivot(index='housing_median_age',
         columns=['total_rooms', 'total_bedrooms'],
         values='median_house_value')
```

total_rooms	5612.0	7650.0	720.0	1501.0	1454.0
total_bedrooms	1283.0	1901.0	174.0	337.0	326.0
housing_median_age					
14.0	NaN	NaN	NaN	73400.0	NaN
15.0	66900.0	NaN	NaN	NaN	NaN
17.0	NaN	NaN	85700.0	NaN	NaN
19.0	NaN	80100.0	NaN	NaN	NaN
20.0	NaN	NaN	NaN	NaN	65500.0

[index, columns, total_bedrooms] combinations must be unique

pivot_table

```
df.pivot_table(index='housing_median_age',
        columns=['total_rooms', 'total_bedrooms'],
        values='median_house_value',
        aggfunc=np.mean)
```

total_rooms	720.0	1454.0	1501.0	5612.0	7650.0
total_bedrooms	174.0	326.0	337.0	1283.0	1901.0
housing_median_age					
14.0	NaN	NaN	73400.0	NaN	NaN
15.0	NaN	NaN	NaN	66900.0	NaN
17.0	85700.0	NaN	NaN	NaN	NaN
19.0	NaN	NaN	NaN	NaN	80100.0
20.0	NaN	65500.0	NaN	NaN	NaN

Data Frame Manipulation

- `df.concat()  # axis=0`

```
df1 = pd.DataFrame({
    "X" : ['A', 'B', 'O', 'AB'],
    "Y" : [15, 12, 10, 12],
    "Z" : [30, 28, 23, 29]},
    index = [1, 2, 3, 4])
df2 = pd.DataFrame({
    "X" : ['O', 'A', 'B'],
    "Y" : [20, 21, 22],
    "Z" : [32, 30, 20],
    "T" : [1, 0, 1]},
    index = [1, 2, 3])

df_new = pd.concat([df1, df2])
```

	X	Y	Z	T
1	A	15	30	NaN
2	B	12	28	NaN
3	O	10	23	NaN
4	AB	12	29	NaN
1	O	20	32	1.0
2	A	21	30	0.0
3	B	22	20	1.0

Data Frame Manipulation

- `df.concat()  # axis=1`

```
df1 = pd.DataFrame({
    "X" : ['A', 'B', 'O', 'AB'],
    "Y" : [15, 12, 10, 12],
    "Z" : [30, 28, 23, 29]},
    index = [1, 2, 3, 4])

df2 = pd.DataFrame({
    "U" : [0, 1, 0],
    "V" : [20, 1, 6]},
    index = [1, 2, 3])

pd.concat([df1, df2], axis=1)
```

	X	Y	Z	U	V
1	A	15	30	0.0	20.0
2	B	12	28	1.0	1.0
3	O	10	23	0.0	6.0
4	AB	12	29	NaN	NaN

Data Frame Manipulation

- `df.fillna(<value>)`

```
df_new.fillna(-1)
```

	X	Y	Z	T
1	A	15	30	NaN
2	B	12	28	NaN
3	O	10	23	NaN
4	AB	12	29	NaN
1	O	20	32	1.0
2	A	21	30	0.0
3	B	22	20	1.0

	X	Y	Z	T
1	A	15	30	-1.0
2	B	12	28	-1.0
3	O	10	23	-1.0
4	AB	12	29	-1.0
1	O	20	32	1.0
2	A	21	30	0.0
3	B	22	20	1.0

Data Frame Manipulation

- Select a subset

```
sub_df = df1[df1.Y > 10]
```

	X	Y	Z
1	A	15	30
2	B	12	28
3	O	10	23
4	AB	12	29

	X	Y	Z
1	A	15	30
2	B	12	28
4	AB	12	29

Data Frame Manipulation

- Select a subset

```
sub_df = df1[df1.X.isin(['AB','A'])]
```

	X	Y	Z
1	A	15	30
2	B	12	28
3	O	10	23
4	AB	12	29

	X	Y	Z
1	A	15	30
4	AB	12	29

Data Frame Manipulation

- Select a subset by columns

```
columns = df1[['X', 'Z']]
```

	X	Y	Z
1	A	15	30
2	B	12	28
3	O	10	23
4	AB	12	29

	X	Z
1	A	30
2	B	28
3	O	23
4	AB	29

Data Frame Manipulation

- Add a new column

```
stocks['value'] = stocks.close*stocks.volume
```

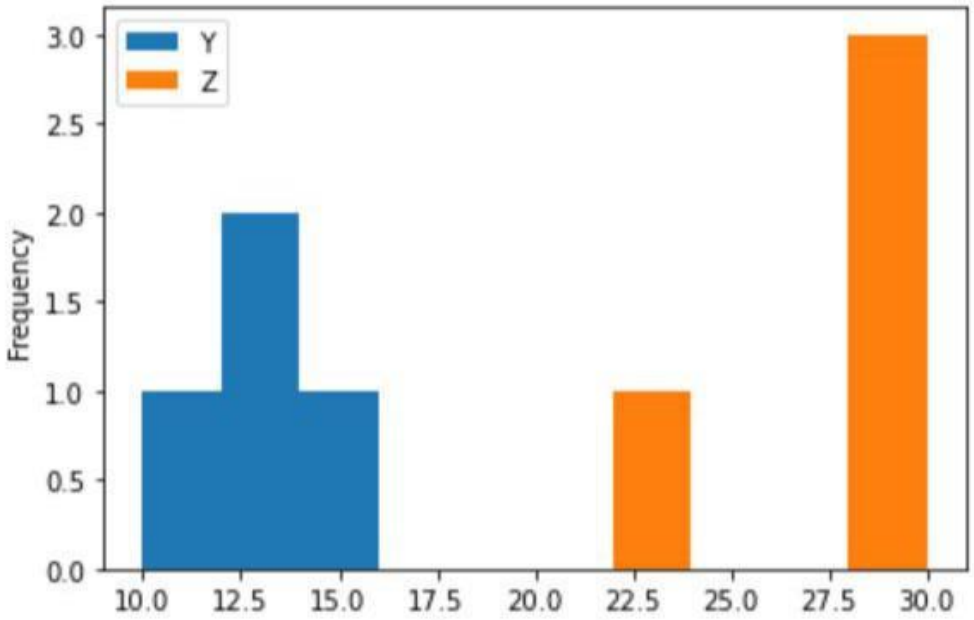
	date	symbol	open	high	low	close	volume	value
0	2019-03-01	AMZN	1655.13	1674.26	1651.00	1671.73	4974877	8.316651e+09
1	2019-03-04	AMZN	1685.00	1709.43	1674.36	1696.17	6167358	1.046089e+10
2	2019-03-05	AMZN	1702.95	1707.80	1689.01	1692.43	3681522	6.230718e+09
3	2019-03-06	AMZN	1695.97	1697.75	1668.28	1668.95	3996001	6.669126e+09
4	2019-03-07	AMZN	1667.37	1669.75	1620.51	1625.95	4957017	8.059862e+09

Data Frame Manipulation

- Plot a chart

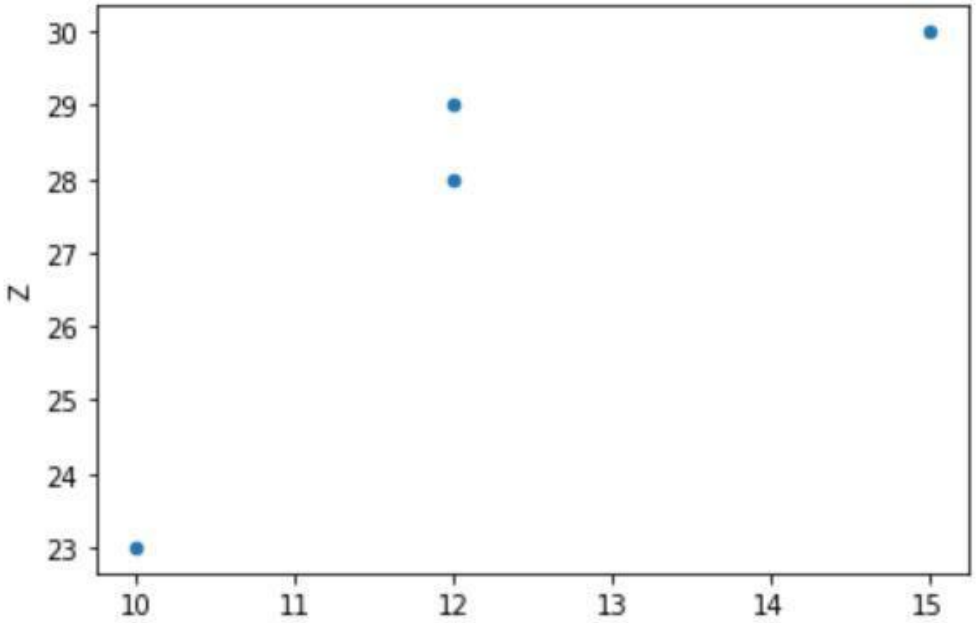
```
df.plot.hist()
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x2545b11a7c0>
```



Data Frame Manipulation

- Plot a chart

```
df.plot.scatter(x='Y', y='Z')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x2545b1bc730>
```